

Shyamoli Textile Engineering College  
Department of Computer Science and Engineering  
**CSE-4237: Digital Image Processing**

**Lec-4: Image Enhancement in the Spatial Domain**

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-3)

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## 4. Histogram of a Digital Image

A histogram represents the *distribution of pixel intensities* (gray levels) within the image.

Let *L* be the total number of possible gray levels in the image (  $L = 2^k$  for a  $k$ -bit image).

Each gray level  $r_k$  has a corresponding count  $n_k$ , which is the number of pixels in the image with that specific gray level.

The histogram  $h(r_k)$  is defined as: 
$$h(r_k) = n_k$$

$r_k$  is the  $k$ -th gray level.

$n_k$  is the number of pixels with intensity  $r_k$ .

## 4.1 Normalized Histogram

To normalize the histogram, each count  $n_k$  is divided by the total number of pixels in the image,  $n$ :

$$p(r_k) = \frac{n_k}{n}$$

$p(r_k)$  is the normalized histogram value for gray level  $r_k$ , representing the probability of occurrence of  $r_k$ .

$n$  is the total number of pixels in the image.

## 4.1.1 Properties of the Normalized Histogram

### Probability Interpretation:

$p(r_k)$  is an estimate of the probability of the gray level  $r_k$  occurring in the image.

For example, if  $p(r_k)=0.2$ , it means 20% of the image's pixels have intensity  $r_k$ .

### Sum of Probabilities:

The sum of all  $p(r_k)$  values equals 1:

$$\sum_{k=0}^{L-1} p(r_k) = 1$$

This is because every pixel in the image has some intensity, and the normalized histogram accounts for all of them.

### 4.1.2 Example of normalized histogram

Suppose we have a grayscale image with  $L = 4$  gray levels

$$(r_0 = 0, r_1 = 1, r_2 = 2, r_3 = 3):$$

Pixel intensities in the image:

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 \\ 3 & 3 & 3 & 2 \end{bmatrix}$$

Show that the sum of all  $p(r_k)$  values equals 1: 
$$\sum_{k=0}^{L-1} p(r_k) = 1$$

**Solution:**

Total number of pixels ( $n$ ):  $4 \times 4 = 16$ .

Histogram  $h(r_k)$ :  $r_0 = 0: n_0 = 3$  (3 pixels have intensity 0).

$$r_1 = 1: n_1 = 4.$$

$$r_2 = 2: n_2 = 4.$$

$$r_3 = 3: n_3 = 5.$$

Normalized histogram  $p(r_k)$ :  $p(r_0) = \frac{n_0}{n} = \frac{3}{16} \approx 0.1875.$

$$p(r_k) = \frac{n_k}{n}$$

$$p(r_1) = \frac{n_1}{n} = \frac{4}{16} = 0.25.$$

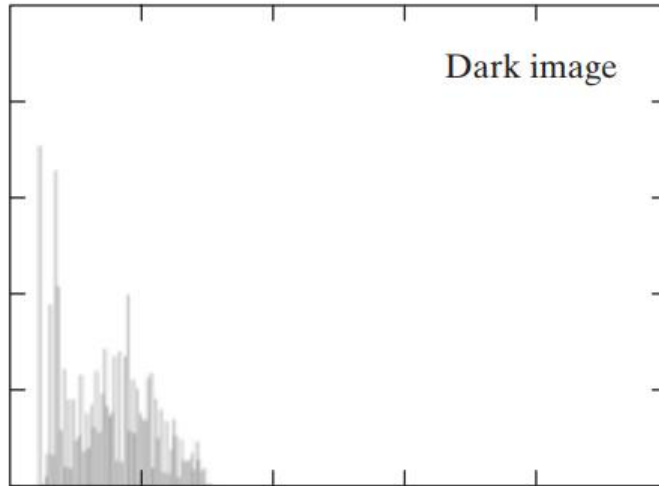
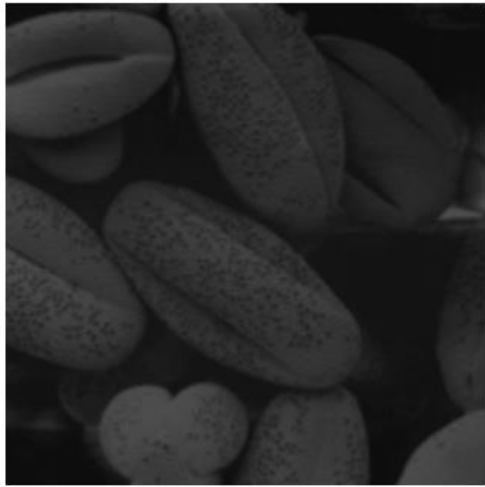
$$p(r_2) = \frac{n_2}{n} = \frac{4}{16} = 0.25.$$

$$p(r_3) = \frac{n_3}{n} = \frac{5}{16} \approx 0.3125.$$

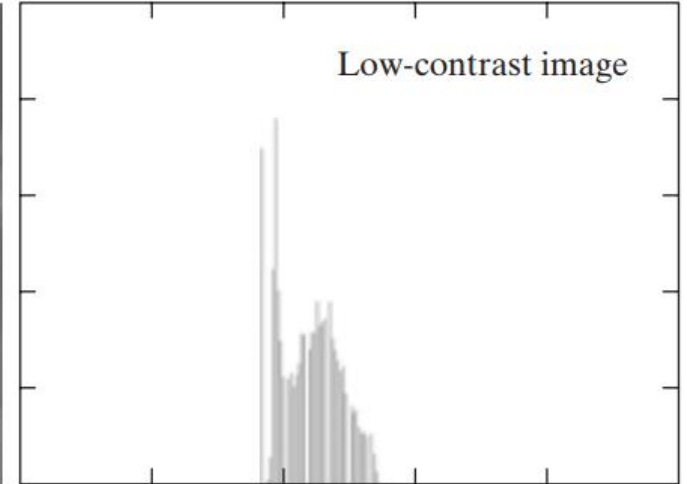
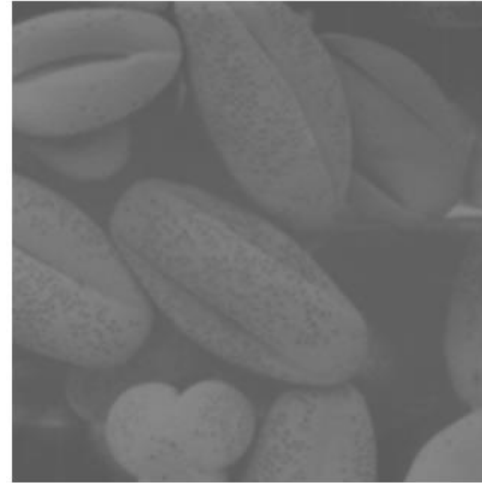
Sum of  $p(r_k)$ :  $0.1875 + 0.25 + 0.25 + 0.3125 = 1$

The horizontal axis of each histogram plot corresponds to gray level values,  $r_k$ .

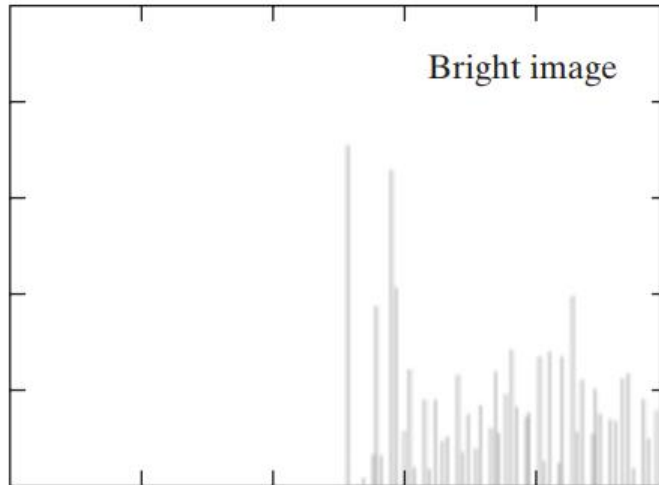
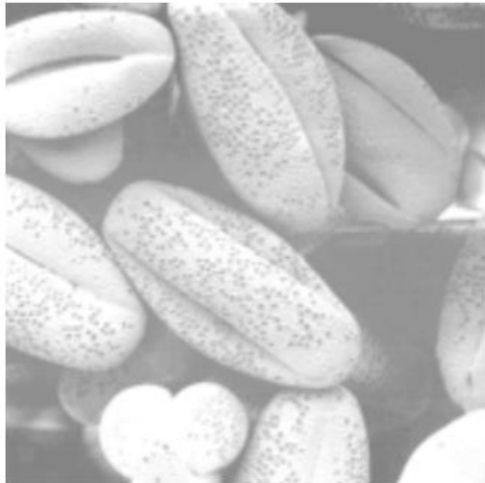
The vertical axis corresponds to values of  $h(r_k) = n_k$  or  $p(r_k) = n_k/n$  if the values are normalized.



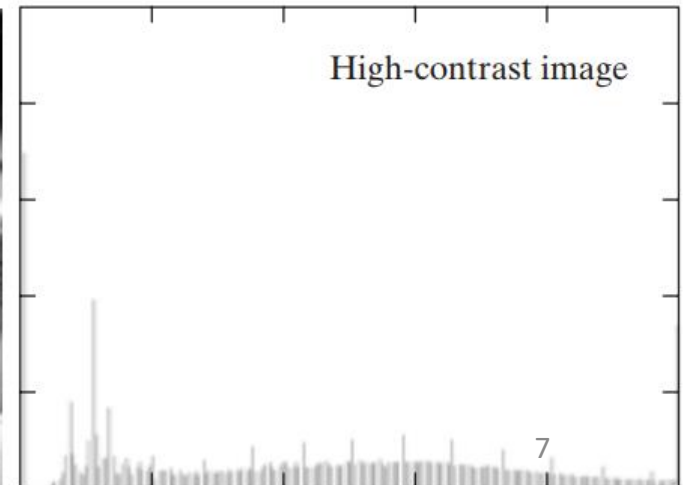
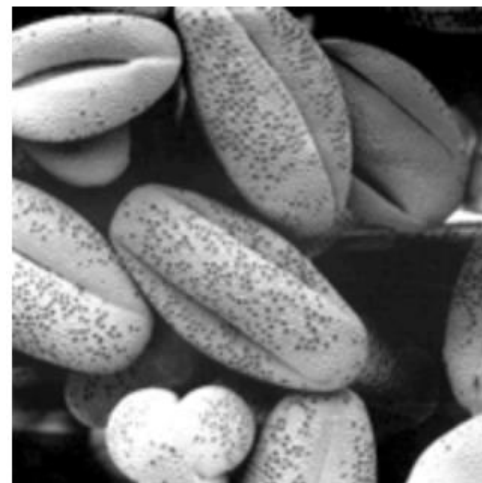
Dark image



Low-contrast image



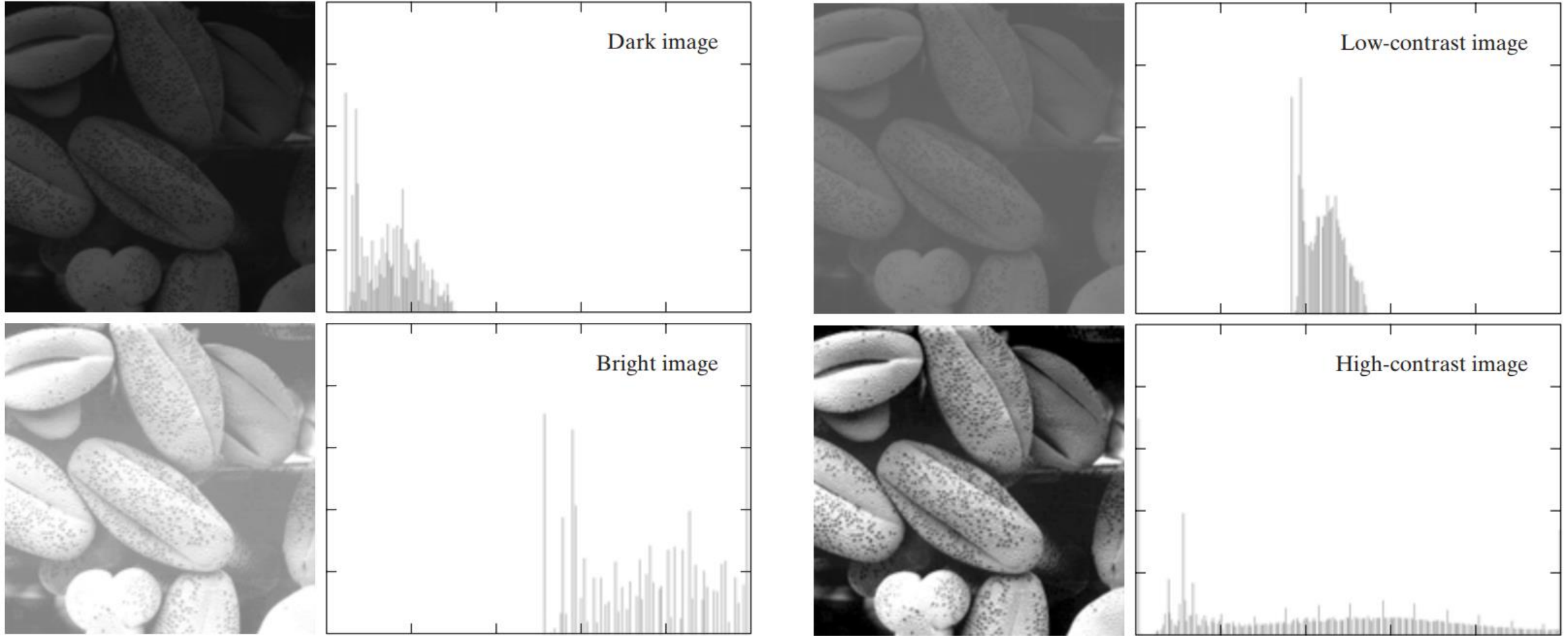
Bright image



High-contrast image

These histogram plots are simply plots of

$$h(r_k) = n_k \text{ versus } r_k \text{ or } p(r_k) = n_k/n \text{ versus } r_k$$



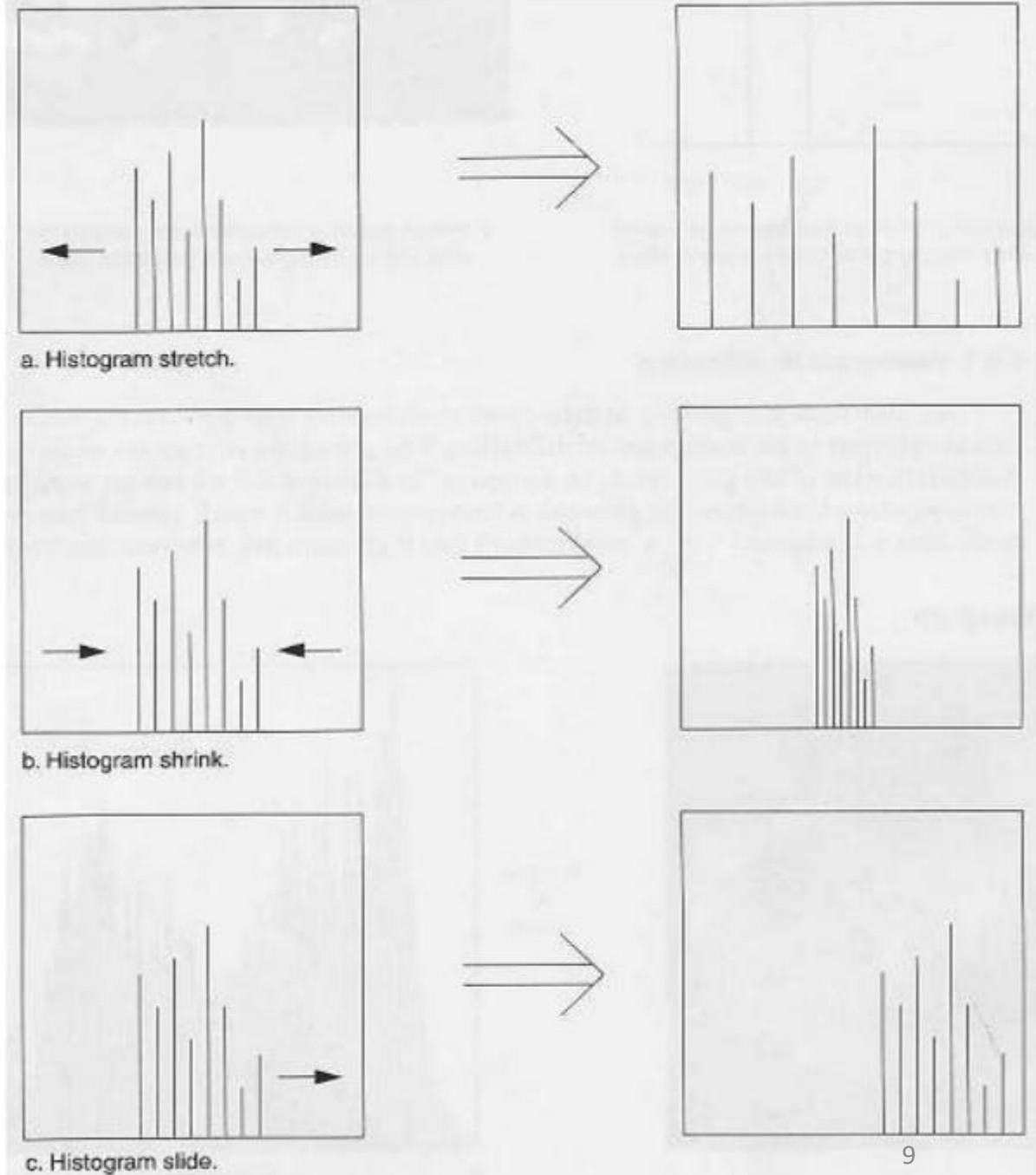
Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms.

## 4.2 Histogram Modifications

The gray level histogram of an image is the distribution of the gray level in an image.

The histogram can be modified by mapping functions, which will

- ❑ Stretch,
- ❑ Shrink (compress),
- ❑ Slide the histogram



## 4.2.1 Stretch the Histogram

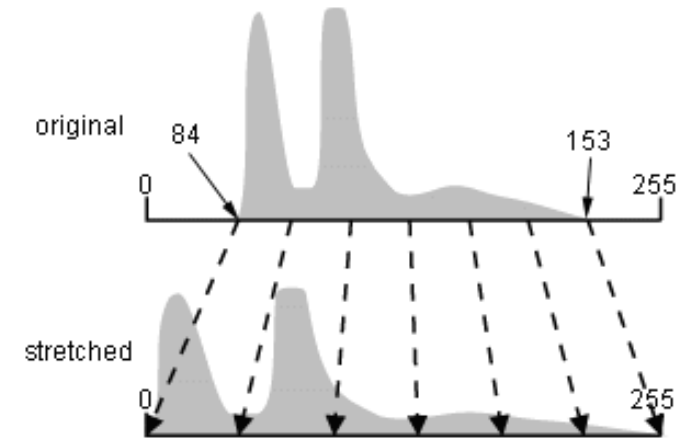
Stretching the histogram increases the contrast by expanding the range of intensity levels.

Mapping Function:  $s = a \cdot (r - r_{\min})$

$r$  = input gray level.

$r_{\min}$  = minimum gray level in the input image.

$a = \frac{s_{\max}}{r_{\max} - r_{\min}}$ , where  $r_{\max}$  and  $s_{\max}$  are the maximum gray levels in the input and output images, respectively.



**Effect:** Stretches the histogram to span the full intensity range ( $[s_{\min}, s_{\max}]$ ).

## 4.2.2 Shrink (Compress) the Histogram

Shrinking compresses the intensity range, effectively reducing contrast.

Mapping Function:  $s = a \cdot r + b$

$$a = \frac{s_{\max} - s_{\min}}{r_{\max} - r_{\min}}.$$

$$b = s_{\min} - a \cdot r_{\min}.$$

**Effect:** Compresses the histogram into a smaller range ( $[s_{\min}, s_{\max}]$ ).

### 4.2.3 Slide the Histogram

Sliding shifts the histogram to the left or right without changing its shape.

Mapping Function:  $s = r + \Delta$

$r$  = input gray level.

$\Delta$  = constant offset (positive for right shift, negative for left shift).

**Effect:** Shifts the histogram by a constant value  $\Delta$ .

## 4.3 Histogram Equalization

Histogram equalization is a technique used in image processing to enhance the contrast of an image by redistributing its gray levels more evenly across the intensity range.

Let the variable  $r$  represent the gray levels of the image to be enhanced.

- ❑ In the initial part of our discussion, we assume that  $r$  has been normalized to the interval  $[0, 1]$ , with  $r = 0$  representing black and  $r = 1$  representing white.
- ❑ Later, we consider a discrete formulation and allow pixel values to be in the interval  $[0, L-1]$ .

**Example:** Assume we have an image with 8 gray levels (from 0 to 7) and the following frequency distribution of pixels:

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 6 |
| 3 | 0 | 5 | 2 |
| 2 | 5 | 1 | 0 |
| 1 | 7 | 2 | 5 |
| 2 | 3 | 7 | 3 |

$n_0 = 2$  pixels with gray level 0

$n_1 = 3$  pixels with gray level 1

$n_2 = 5$  pixels with gray level 2

$n_3 = 4$  pixels with gray level 3

$n_4 = 0$  pixels with gray level 4

$n_5 = 3$  pixels with gray level 5

$n_6 = 1$  pixel with gray level 6

$n_7 = 2$  pixels with gray level 7

## Step 1: Calculate Normalized Histogram

$$p_r(r_k) = \frac{n_k}{n} \quad k = 0, 1, 2, \dots, L - 1$$

First, we normalize the histogram by dividing the count of each gray level by the total number of pixels, which is  $n=20$ .

So, we calculate

$$p(r_k) = \frac{n_k}{n}:$$

| Gray Level $r_k$ | Count $n_k$ | Normalized Histogram $p(r_k) = \frac{n_k}{n}$ |
|------------------|-------------|-----------------------------------------------|
| 0                | 2           | $\frac{2}{20} = 0.1$                          |
| 1                | 3           | $\frac{3}{20} = 0.15$                         |
| 2                | 5           | $\frac{5}{20} = 0.25$                         |
| 3                | 4           | $\frac{4}{20} = 0.2$                          |
| 4                | 0           | $\frac{0}{20} = 0.0$                          |
| 5                | 3           | $\frac{3}{20} = 0.15$                         |
| 6                | 1           | $\frac{1}{20} = 0.05$                         |
| 7                | 2           | $\frac{2}{20} = 0.1$                          |

## Step 2: Calculate Cumulative Distribution Function (CDF) $T(r_k) = \sum_{j=0}^k p(r_j)$

Now, we compute the cumulative sum of the normalized histogram. This gives us the CDF  $T(r_k)$ .

| Gray Level $r_k$ | $p(r_k)$ | Cumulative Distribution Function $T(r_k)$ |
|------------------|----------|-------------------------------------------|
| 0                | 0.1      | 0.1                                       |
| 1                | 0.15     | $0.1 + 0.15 = 0.25$                       |
| 2                | 0.25     | $0.25 + 0.25 = 0.5$                       |
| 3                | 0.2      | $0.5 + 0.2 = 0.7$                         |
| 4                | 0.0      | $0.7 + 0.0 = 0.7$                         |
| 5                | 0.15     | $0.7 + 0.15 = 0.85$                       |
| 6                | 0.05     | $0.85 + 0.05 = 0.9$                       |
| 7                | 0.1      | $0.9 + 0.1 = 1.0$                         |

### Step 3: Calculate the Equalized Gray Levels $s_k$

Finally, we map the cumulative distribution to the output intensity levels. Since we have  $L=8$  gray levels (from 0 to 7), we compute the new gray level  $s_k$  for each  $r_k$  using the formula:

$$s_k = (L - 1) \cdot T(r_k)$$

| Gray Level $r_k$ | CDF $T(r_k)$ | Transformed Gray Level $s_k = (L - 1) \cdot T(r_k)$ |
|------------------|--------------|-----------------------------------------------------|
| 0                | 0.1          | $(8 - 1) \cdot 0.1 = 0.7$                           |
| 1                | 0.25         | $(8 - 1) \cdot 0.25 = 1.75$                         |
| 2                | 0.5          | $(8 - 1) \cdot 0.5 = 3.5$                           |
| 3                | 0.7          | $(8 - 1) \cdot 0.7 = 4.9$                           |
| 4                | 0.7          | $(8 - 1) \cdot 0.7 = 4.9$                           |
| 5                | 0.85         | $(8 - 1) \cdot 0.85 = 5.95$                         |
| 6                | 0.9          | $(8 - 1) \cdot 0.9 = 6.3$                           |
| 7                | 1.0          | $(8 - 1) \cdot 1.0 = 7.0$                           |

Since the result is not an integer, we round it to the nearest integer.

| Gray Level $r_k$ | $s_k$ (Rounded) |
|------------------|-----------------|
| 0                | 1               |
| 1                | 2               |
| 2                | 4               |
| 3                | 5               |
| 4                | 5               |
| 5                | 6               |
| 6                | 6               |
| 7                | 7               |

| Original Gray Level $r_k$ | Number of pixels | Histogram Equalized Gray level $s_k$ |
|---------------------------|------------------|--------------------------------------|
| 0                         | 2                | 1                                    |
| 1                         | 3                | 2                                    |
| 2                         | 5                | 4                                    |
| 3                         | 4                | 5                                    |
| 4                         | 0                | 5                                    |
| 5                         | 3                | 6                                    |
| 6                         | 1                | 6                                    |
| 7                         | 2                | 7                                    |

## Step 4: Result

The original gray levels  $r_k$  are mapped to the new gray levels  $s_k$ , which increases the contrast of the image.

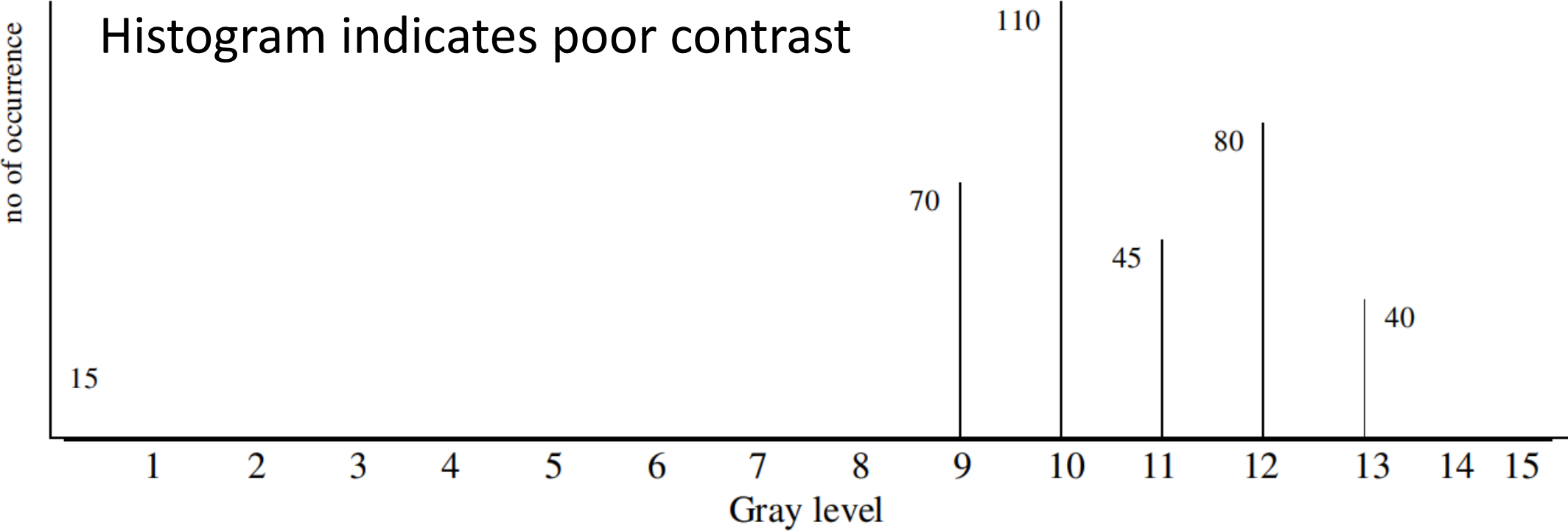
**Original Image:** Pixels are mapped to  $[0,1,2,3,4,5,6,7]$ .

**Equalized Image:** Pixels are mapped to the new levels  $[1,2,4,5,5,6,6,7]$ .

This process spreads the pixel intensities across the available range more evenly, improving the contrast of the image.

**Example:** Suppose a 4-bit greyscale image has the histogram shown in the table below. Draw the histogram graph and make a necessary transformation of gray values.

|                |    |   |   |   |   |   |   |   |   |    |     |    |    |    |    |    |
|----------------|----|---|---|---|---|---|---|---|---|----|-----|----|----|----|----|----|
| Gray-level (i) | 0  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9  | 10  | 11 | 12 | 13 | 14 | 15 |
| n(i)           | 15 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 70 | 110 | 45 | 80 | 40 | 0  | 0  |



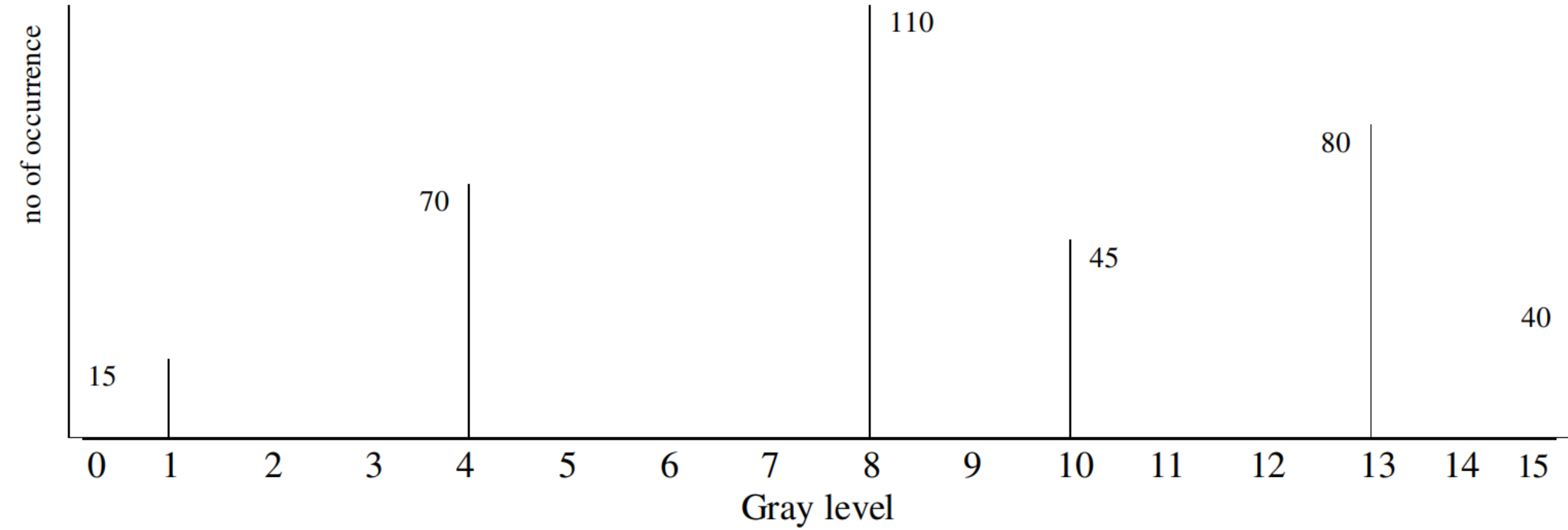
With  $n = 360$  we expect this image to be uniformly bright with few dark dots.

To equalize this histogram, we form running total of the 2 and multiply each by  $15/360 = 1/24$ .

We now have the following transformation of gray values obtain by reading off the first and the last columns in the table below.

| Gray level (i) | $n(i)$ | $\sum n(i)$ | $\frac{15}{360} \sum n(i)$ | Rounded value |
|----------------|--------|-------------|----------------------------|---------------|
| 0              | 15     | 15          | 0.63                       | 1             |
| 1              | 0      | 15          | 0.63                       | 1             |
| 2              | 0      | 15          | 0.63                       | 1             |
| 3              | 0      | 15          | 0.63                       | 1             |
| 4              | 0      | 15          | 0.63                       | 1             |
| 5              | 0      | 15          | 0.63                       | 1             |
| 6              | 0      | 15          | 0.63                       | 1             |
| 7              | 0      | 15          | 0.63                       | 1             |
| 8              | 0      | 15          | 0.63                       | 1             |
| 9              | 70     | 85          | 3.65                       | 4             |
| 10             | 110    | 195         | 8.13                       | 8             |
| 11             | 45     | 240         | 10                         | 10            |
| 12             | 80     | 320         | 13.33                      | 13            |
| 13             | 40     | 360         | 15                         | 15            |
| 14             | 0      | 360         | 15                         | 15            |
| 15             | 0      | 360         | 15                         | 15            |

# Histogram after equalization



**Example:** We have an image with 3 bit /pixel, so the possible range of values is 0 to 7 . We have an image with the following histogram

| Gray level value             | 0  | 1 | 2 | 3 | 4  | 5 | 6 | 7 |
|------------------------------|----|---|---|---|----|---|---|---|
| No. of Pixel histogram value | 10 | 8 | 9 | 2 | 14 | 1 | 5 | 2 |



### Step 1:

Great a running sum of histogram values. This means that the **first values** is **10**, The **second** is  **$10+8=18$** , next is  **$10+8+9=27$** , and soon. Here we get **10,18,29,43,44,49,51**.

| Gray level value | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|------------------|----|----|----|----|----|----|----|----|
| No. of Pixel     | 10 | 8  | 9  | 2  | 14 | 1  | 5  | 2  |
| Run sum          | 10 | 18 | 27 | 29 | 43 | 44 | 49 | 51 |

### Step 2:

Normalize by dividing by total number of pixels. The total number of pixels is  **$10+8+9+2+14+1+5+2=51$** .

| Gray level value | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
|------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| No. of Pixel     | 10    | 8     | 9     | 2     | 14    | 1     | 5     | 2     |
| Run sum          | 10    | 18    | 27    | 29    | 43    | 44    | 49    | 51    |
| Normalize        | 10/51 | 18/51 | 27/51 | 29/51 | 43/51 | 44/51 | 49/51 | 51/51 |

### Step 3:

Multiply these values by the maximum gray level values in this case 7 , and then round the result to the closet integer. After this is done we obtain 1,2,4,4,6,6,7,7.

| Gray level value | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
|------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| No. of Pixel     | 10    | 8     | 9     | 2     | 14    | 1     | 5     | 2     |
| Run sum          | 10    | 18    | 27    | 29    | 43    | 44    | 49    | 51    |
| Normalize        | 10/51 | 18/51 | 27/51 | 29/51 | 43/51 | 44/51 | 49/51 | 51/51 |
| Multiply by 7    | 1     | 2     | 4     | 4     | 6     | 6     | 7     | 7     |

### Step 4:

Map the original values to the results from step 3 by a one-to-one correspondence.

| Old | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|-----|---|---|---|---|---|---|---|---|
| New | 1 | 2 | 4 | 4 | 6 | 6 | 7 | 7 |

All pixel in the original image with gray level 0 are set to 1, values of 1 are set to 2, and values 2 set to 4, and values 3 set to 4, and so on.

|            |          |          |          |          |          |          |          |          |
|------------|----------|----------|----------|----------|----------|----------|----------|----------|
| <b>Old</b> | <b>0</b> | <b>1</b> | <b>2</b> | <b>3</b> | <b>4</b> | <b>5</b> | <b>6</b> | <b>7</b> |
| <b>New</b> | <b>1</b> | <b>2</b> | <b>4</b> | <b>4</b> | <b>6</b> | <b>6</b> | <b>7</b> | <b>7</b> |

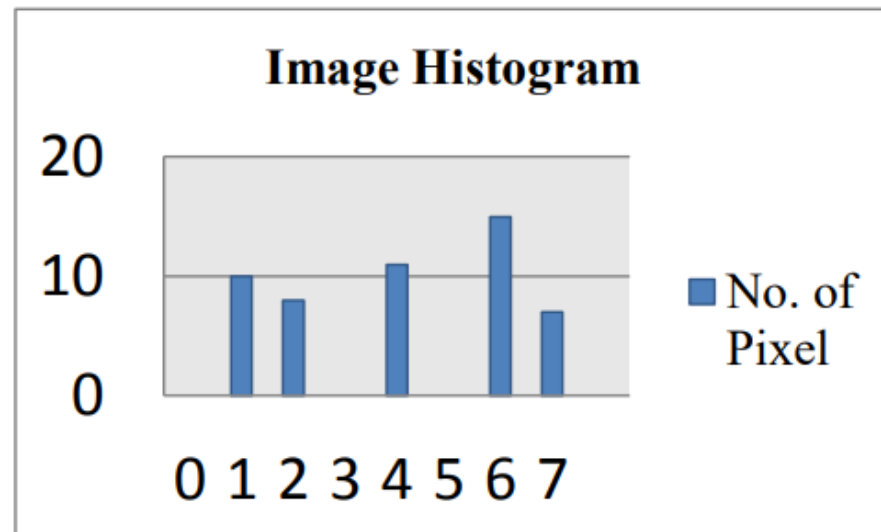
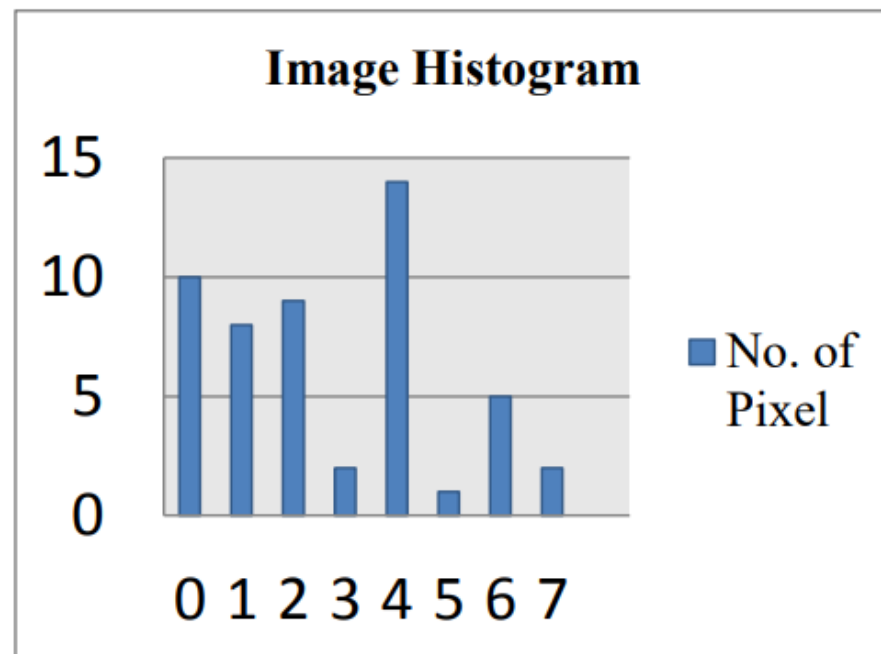
**Before  
Histogram  
Equalization**

|                                             |           |          |          |          |           |          |          |          |
|---------------------------------------------|-----------|----------|----------|----------|-----------|----------|----------|----------|
| <b>Gray level<br/>value</b>                 | <b>0</b>  | <b>1</b> | <b>2</b> | <b>3</b> | <b>4</b>  | <b>5</b> | <b>6</b> | <b>7</b> |
| <b>No. of Pixel<br/>histogram<br/>value</b> | <b>10</b> | <b>8</b> | <b>9</b> | <b>2</b> | <b>14</b> | <b>1</b> | <b>5</b> | <b>2</b> |

**After  
Histogram  
Equalization**

|                                             |          |           |          |          |           |          |           |          |
|---------------------------------------------|----------|-----------|----------|----------|-----------|----------|-----------|----------|
| <b>Gray level<br/>value</b>                 | <b>0</b> | <b>1</b>  | <b>2</b> | <b>3</b> | <b>4</b>  | <b>5</b> | <b>6</b>  | <b>7</b> |
| <b>No. of Pixel<br/>histogram<br/>value</b> | <b>0</b> | <b>10</b> | <b>8</b> | <b>0</b> | <b>11</b> | <b>0</b> | <b>15</b> | <b>7</b> |

**Before Histogram Equalization**



**After Histogram Equalization**



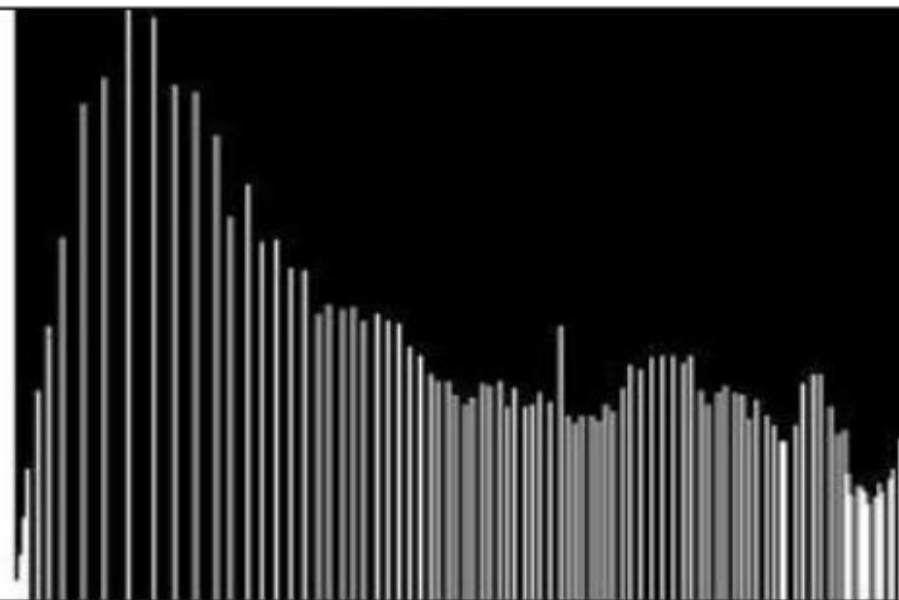
**Original image**



**Histogram of original image**



**Image after histogram equalization**



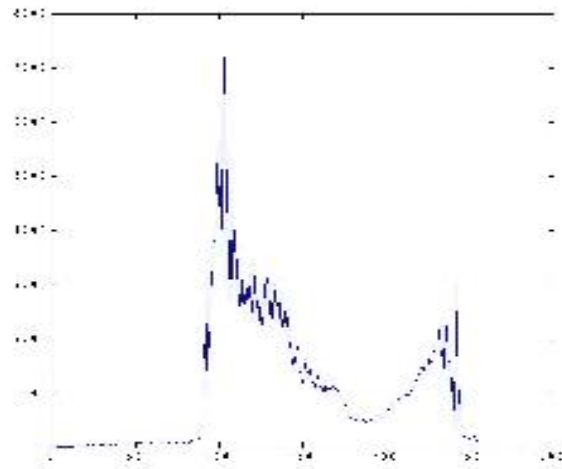
**Histogram after equalization**



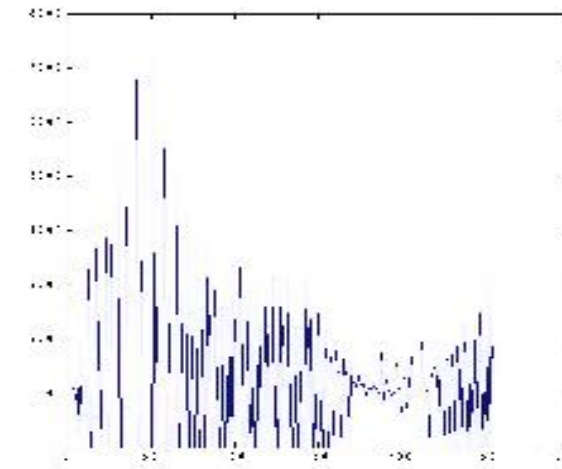
Original Image



After Histogram Equalisation



Original Histogram



Histogram after equalisation

